

One Thing, Doing Everything

A TED-Style Public Lecture on Unified Substrate Theory

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THE STRANGEST QUESTION I KNOW

I want to start with something I'd like you to feel before I explain. Not understand — feel.

Right now, there are photons hitting your eyes. Billions of them per second, every second, streaming from the lights in this room and bouncing off the faces of everyone around you. Those photons left their sources — the lamps, the screens, the sun — and traveled through what we call empty space to get to you.

At the same time, every atom in your body — every one of the roughly seven-octillion atoms that make up the thing you call yourself — is vibrating. Not because you're nervous, though maybe you are. But because that's what atoms do. They don't sit still. Ever. The electrons around their nuclei exist in a kind of permanent quantum fidget, smeared across possible locations like a fog, collapsing into a definite spot only when something forces them to commit.

Meanwhile, the chair beneath you is bending spacetime. I know that sounds absurd. But it's true. Any object with mass — including you, including your chair, including the coffee you might be holding — curves the fabric of the

universe around it, just slightly, just a little. That curvature is what we call gravity.

So we have: photons streaming through space. Quantum matter fizzing with uncertainty. Spacetime bending under weight. Three things. Three wildly different-sounding phenomena.

"What if I told you they are all the same thing — just one field, one underlying reality, caught doing different things at different scales?"

That's the idea I want to share with you today. It has a name: Unified Substrate Theory. And whether or not I've gotten every detail right — and I'll be honest with you about where the gaps are — I believe it is pointing at something real. Something that physics has been circling for a hundred years without quite landing on.

I'm not a professor. I don't have a lab or a university affiliation or a grant from anyone. I'm a guy from Walstonburg, North Carolina, who got obsessed with the deepest question in science and couldn't let it go. And maybe that's exactly the kind of person who's allowed to say something simple that the experts have been too close to see.

Or maybe I'm wrong. Either way — let's take the ride.

THE TWO PERFECT THEORIES THAT CAN'T BE FRIENDS

Here is the central crisis of modern physics, and I want you to really feel how strange it is.

We have two theories of how the universe works. Both of them are correct. Both of them have been tested to extraordinary precision — more precisely than almost anything else in all of science. And they flatly contradict each other.

The first theory is **General Relativity**. Einstein's masterpiece, finished in 1915. It describes gravity — how mass warps space and time, how planets orbit stars, how light bends around galaxies, how clocks tick differently depending on where they are in a gravitational field. It works so well that GPS satellites have to correct for relativistic time dilation every single day, or your navigation would drift by miles within hours. General Relativity treats spacetime as a smooth, continuous fabric that stretches and curves.

The second theory is **Quantum Mechanics**. Developed in the early twentieth century, refined ever since. It describes how matter and energy behave at the tiniest scales — atoms, electrons, photons, quarks. It works so well that the computer you used this morning, the MRI machine your doctor might have used on you, the laser in every barcode scanner — all of them run on quantum mechanics. Quantum theory treats the world as fundamentally granular: discrete packets, probabilistic outcomes, particles that don't have definite positions until you measure them.

"Imagine you have two maps of the same city. One is

incredibly accurate for the highway system. The other is incredibly accurate for the alleyways. But when you try to overlap them, they don't match — the streets are in different places, the scale is wrong, and the coordinate systems seem to describe different cities altogether."

That is exactly the situation physicists are in. General Relativity describes a smooth, deterministic, continuous universe. Quantum Mechanics describes a grainy, probabilistic, discrete one. When you try to apply both theories to the same situation — say, the interior of a black hole, or the very first moment of the Big Bang — you get nonsense. Infinities where there should be finite numbers. Predictions that are meaningless. The math breaks down.

This is not a minor technical problem. This is a gaping hole in our understanding of reality. For over a century, the greatest minds in physics have been trying to build a bridge between these two frameworks. And here's what makes it even more maddening: the bridge has to exist. There is only one universe. There cannot, in the end, be two incompatible sets of rules running simultaneously. Somewhere, underneath both theories, there has to be one deeper truth.

Finding it is the biggest open question in science.

THE BRAVE ATTEMPTS, AND THE WALLS THEY HIT

I want to be fair here, because the people who have worked on this problem are brilliant, and they haven't been wasting their time.

The most famous attempt is **String Theory**. The core idea is genuinely beautiful: instead of treating fundamental particles as point-like objects — tiny dots of matter — what if they are actually tiny, vibrating loops of energy? Like guitar strings, but unimaginably small, a hundred billion billion times smaller than a proton. Different vibrational modes of the string would produce different particles: one mode is an electron, another is a quark, another is a graviton — the hypothetical particle that would carry the force of gravity.

String theory naturally produces a graviton. That's a real achievement. And the mathematics that has come out of string research has been genuinely profound — it's contributed to pure mathematics, to our understanding of black holes, to many corners of theoretical physics. But after decades of work, string theory has a serious problem: it makes essentially no predictions that we can test with current experiments. Worse, it seems to generate not one universe but something like ten to the five-hundredth power of possible universes — an almost incomprehensible number of theoretical possibilities, with no way to choose which one is ours.

The other major contender is **Loop Quantum Gravity**. This approach takes a different starting point: instead of adding extra dimensions or new objects, what if spacetime itself is quantized? What if, at the very smallest scale, space is not a smooth continuum but a kind of discrete network — a fabric woven from tiny loops, each roughly the Planck length in size, which is about ten to the minus thirty-five meters? Spacetime, in this picture, is not infinitely divisible. There's a minimum unit of space, like a pixel on a screen but unimaginably finer.

Loop Quantum Gravity has also produced real insights. It elegantly resolves some of the singularity problems — it suggests the Big Bang was not an infinitely dense point but a "bounce" from a previous collapsing phase. But it has its own trouble: it hasn't yet been fully reconciled with quantum field theory, and it, too, struggles to produce testable predictions at energies we can access.

The Honest Assessment

Both programs are serious science, done by serious people. Neither is nonsense. But after decades, neither has cracked it. The unification problem remains open — and that means the door is still unlocked for a new idea.

ONE FIELD. ONE REALITY. EVERYTHING ELSE IS A RIPPLE.

All right. Here is the idea. I'm going to walk you through it slowly, because I want it to actually land — not just wash over you and sound vaguely impressive.

The central claim of Unified Substrate Theory is this:

"There is one field. It fills everything. It IS everything. We call it the substrate. And every phenomenon in the universe — matter, energy, spacetime, gravity, quantum behavior — is

this one field in different states."

Let me build that up piece by piece.

Spacetime Is the Substrate at Rest

In standard physics, spacetime is usually described as the stage on which the play of the universe happens. Particles are the actors. Forces are the interactions between actors. The stage itself is just... there. Passive.

The substrate framework rejects that picture entirely. There is no empty stage. The substrate is not a background — it's the entire show. What we call "empty spacetime" — the vacuum, the void between galaxies — is the substrate in its lowest-energy, most settled state. Like a pond on a windless day. Flat. Quiet. But absolutely present.

Think of the ocean. When it's calm — when there are no storms, no disturbances — you have just water. It looks like nothing is happening. But the water is still there. The potential for waves is still there. The calm is not the absence of the ocean. The calm is the ocean being itself.

That's spacetime. The substrate, undisturbed.

Particles Are Knots in the Field

Now here's where it gets really interesting. If the substrate is like that ocean, what is matter? What is an electron, a proton, a quark?

The answer, in this framework, is topology. Specifically: knots.

Take a rope. Tie a knot in it. Now here's a question: can you remove that knot by sliding it off the end of the rope? No — not without untying it. The

knot is a stable, persistent structure. It has properties: a shape, a handedness (you can tie a knot left-handed or right-handed, and those are genuinely different), a complexity. And — critically — if you stretch the rope or move it around, the knot moves with it. The knot is not separate from the rope. It is the rope, organized in a particular way.

In Unified Substrate Theory, fundamental particles are stable topological knots in the substrate. An electron is a specific type of knot. A quark is a different type. A photon is not a knot at all — it's a wave, a ripple propagating through the medium.

This explains something that has bothered physicists for decades: why is every electron in the universe *identical*? Not just similar — literally identical. The same mass, the same charge, the same spin, everywhere in the observable universe, across billions of light-years. If electrons were manufactured objects, you'd expect some variation, some defects. But they're perfect copies.

With the knot model, this makes immediate sense. A specific knot type has specific properties determined by its topology — its mathematical structure — not by where it was "made" or what it's made of. Every trefoil knot is identical to every other trefoil knot. The electron's identity is encoded in the structure of the twist itself, not in any material it's composed of.

Forces Are Pressure Gradients

What about forces — electromagnetism, the strong nuclear force, the weak nuclear force? In the substrate framework, these are pressure gradients: regions where the substrate is more compressed, more stretched, or more twisted than its surroundings. The knots distort the medium around them, and other knots feel that distortion and respond to it. What we experience as

a "force" between two particles is the substrate mediating a change in one knot's configuration due to another knot's influence on the medium between them.

Imagine two whirlpools in a bathtub. They don't touch each other. But if you drop them both into the same tub, the water circulating around one will affect the water flowing near the other. They'll drift toward each other, or push apart, depending on which way they're spinning. They interact not by reaching out and grabbing each other — but through the medium they share.

That's how forces work in the substrate picture. No messenger particles flying back and forth. Just the field, communicating with itself.

Quantum Behavior Is How the Field Works at Tiny Scales

Here is one of the most beautiful parts of the theory, and it is the part that I find most personally compelling.

Quantum mechanics is famously weird. Particles exist in superpositions. Measurements disturb systems. There's an uncertainty principle that says you cannot know both the position and the momentum of a particle exactly — if you nail down one, the other becomes indeterminate. Why? Where does this strangeness come from?

In the substrate framework, the answer is almost obvious once you've accepted the model: the substrate has a finite amount of information per unit volume. It is not infinitely precise. It cannot simultaneously encode, at any given point, both a particle's exact location and its exact motion. That's not a limitation of our measuring instruments — it's a fundamental feature of the medium itself. The field simply cannot hold that much information in one place.

The uncertainty principle isn't a mysterious fact about quantum particles. It's a direct consequence of the substrate's finite resolution. Reality has pixels. Very, very small ones — near the Planck scale — but pixels nonetheless. And you can't read two things from one pixel at the same time.

Gravity Is the Field Curving

And finally: gravity. Here, the substrate framework agrees beautifully with Einstein, but reframes the picture.

The classic image of gravity — which Einstein himself endorsed — is a bowling ball sitting on a rubber sheet. The bowling ball deforms the sheet, and a marble rolled nearby spirals toward it not because it's "attracted" but because it's following the curve of the sheet. Spacetime bends around mass, and things follow the bend.

In the substrate framework, this is exactly right — except the rubber sheet is not a passive backdrop. It IS the substrate. When energy or mass is concentrated in one region — a star, a planet, your body sitting in that chair — the substrate is densely organized there, and this organization creates a gradient in the surrounding field. Things move along that gradient. Gravity is not a force shooting out from the sun to grab the Earth. It's the Earth following the path of least resistance through a curved medium. There is no graviton needed. The curvature is real. And it's made of the same stuff as everything else.

FIVE MYSTERIES THAT SUDDENLY MAKE SENSE

I could sell you a beautiful idea, and it could be beautiful and wrong. Science isn't poetry — you don't get points just for elegance. So let me show you five

places where this framework doesn't just describe what we already know — it illuminates things that were previously dark.

1. Why Every Electron Is Identical

I already touched on this, but let me hammer it home. In conventional particle physics, the fact that all electrons are identical is simply assumed. It's baked into the theory as a postulate. We don't explain it — we just accept it and move on. In the substrate model, it falls out naturally: an electron is a specific topological knot type, and identical knot types have identical properties by mathematical necessity. The theory explains what the old theory just declared.

2. Where the Uncertainty Principle Comes From

Same story. Quantum mechanics gives you the uncertainty principle as a foundational axiom. You're told: this is how it is. Don't ask why. In the substrate model, it's a consequence of the substrate's finite information density at the Planck scale. It's not mysterious anymore — it's actually intuitive. A medium that can't hold infinite precision can't give you infinite precision. That's not a paradox. That's just... how things are when reality is a physical field rather than a mathematical abstraction.

3. Why Quantum Entanglement Isn't Spooky

Einstein called entanglement "spooky action at a distance," and he hated it. Two particles become entangled, and from that point on, measuring one instantly affects the other — no matter how far apart they are. Light-years apart, even. How can measuring something here change something there instantaneously? Nothing is supposed to travel faster than light.

The substrate model offers a clean resolution. When two particles are entangled, they share a real, physical connection in the substrate. They are

not simply "correlated in our knowledge" — they are topologically linked in the field. And from the substrate's perspective, they were never truly far apart. Distance is a feature of the large-scale structure of the field. But at the substrate level — the level below space — these particles maintain a genuine connection, like two knots on the same piece of rope. Measuring one reorganizes the rope, and the other knot instantly knows about it. Not because information traveled between them — but because they were always, at the deepest level, together.

4. What Actually Happened at the Big Bang

In the standard model, the Big Bang is a singularity — a point of infinite density from which everything emerged. Which is... not really a description. It's the place where the physics stops working and we throw up our hands.

The substrate model reframes this. The Big Bang was a phase transition. Think of water freezing into ice. Before the transition, you have a disordered fluid — molecules moving freely, no fixed structure, no long-range order. When the temperature drops below a critical threshold, suddenly the system reorganizes: crystalline structure emerges, long-range patterns appear, the geometry of the medium changes completely.

The substrate before the Big Bang was in a pre-geometric state — not nothing, but a disordered field without the organized structure of space and time. Space and time were not yet defined. Then, at the moment of the Big Bang, the field underwent a phase transition. It reorganized into the structured, geometric state we call our universe. Spacetime didn't emerge from nothing. It emerged from the substrate reorganizing itself, the way ice emerges from water.

5. Why the Universe's Expansion Is Accelerating

In 1998, astronomers discovered that the universe is not just expanding — it's expanding faster and faster. Something is pushing everything apart. We call it dark energy, which is mostly a name for our ignorance. We have no good explanation for where it comes from or why it has the value it does.

Here's the substrate picture: the phase transition of the Big Bang wasn't perfect. It wasn't a clean, instantaneous reorganization. The substrate settled mostly into our spacetime geometry — but not completely. There is a residual tension in the field, a pressure from the slightly-unfinished phase transition, still slowly working itself out. That residual tension is what we observe as the cosmological constant — the energy of "empty" space — and it's what's driving the accelerating expansion. The universe is still, very gently, finishing the process that began 13.8 billion years ago.

REAL EXPERIMENTS. HAPPENING RIGHT NOW.

This is the part that matters most for whether this is science or philosophy. A theory that makes no predictions is not a scientific theory — it's a story. So let me tell you what the substrate framework predicts, and where we are in testing those predictions.

The good news: we don't have to wait. Some of these experiments are already running.

Test 1: The Photon Timing Experiment

If the substrate has structure at the Planck scale — if spacetime is not perfectly smooth but has a kind of fine-grained texture — then different-energy photons should travel through it at very slightly different speeds. The effect would be completely unmeasurable over short distances. But if the photons come from gamma-ray bursts billions of light-years away, those tiny differences in travel time would accumulate over the journey and become detectable on arrival. High-energy photons should arrive fractionally later than low-energy ones. The Fermi Gamma-ray Space Telescope has been collecting exactly this data from gamma-ray bursts for over a decade, and physicists are actively analyzing it for precisely this signature. Recent studies in 2025 examined three remarkable high-energy photons from GRBs and found possible hints of the effect — though more data is needed to reach a firm conclusion.

Test 2: Gravitational Wave Echoes

When two black holes spiral into each other and merge — an event detected by LIGO and Virgo as a ripple in spacetime — the standard prediction is a specific pattern of gravitational waves: a chirp that rises in frequency, peaks at merger, and then rings down to silence. But if the substrate has structure, if black hole horizons are not perfectly smooth surfaces but have some quantum texture, there should be faint, delayed echoes after the main signal. Like sound bouncing off the walls of a room after the original noise is gone. LIGO's teams are actively searching for these echoes in their data. They haven't found them conclusively yet — but they're looking, and the absence of a confirmed detection is already placing constraints on what the substrate's texture can look like.

Test 3: The Quantum Gravity Table-Top Experiment

This one might be my favorite, because it sounds like it shouldn't be possible but actually might be. Take a small weight — a microscopic diamond, say, containing billions of atoms. Put it into a genuine quantum superposition of two positions simultaneously, using the tricks of quantum optics. Place another identical weight nearby. If the two weights become quantum-mechanically entangled with each other through gravity alone — with no electromagnetic interaction between them — then gravity must be fundamentally quantum in nature. It cannot be a classical field. Several experimental groups in Europe and the United States are working toward this experiment right now. The technology is almost there. If it works, it will be the first direct evidence that gravity is quantum mechanical, which is the central assumption underlying the entire substrate framework.

Test 4: The Residual Tension and Dark Energy

The substrate model makes a specific prediction about the nature of dark energy: it should not be a perfectly constant value forever. As the universe's unfinished phase transition continues to settle, the residual tension should evolve — very slowly — over cosmic time. This means the acceleration of the universe should not be perfectly constant across all of cosmic history. Current and upcoming galaxy surveys — including the Dark Energy Spectroscopic Instrument (DESI), which released its first major results in 2024, and the Euclid space telescope, launched in 2023 — are measuring the expansion history of the universe with unprecedented precision, specifically looking for evidence that dark energy has changed over time. Early DESI results showed hints of exactly this kind of evolution. The substrate framework predicts this. We're watching the data come in.

The Scientific Bottom Line

These are not futuristic experiments waiting for technology that doesn't exist yet. Fermi is flying. LIGO is collecting data. Gravity entanglement experiments are in labs. DESI and Euclid are observing. The predictions are on the table. The experiments are running. We will have answers.

THREE STEPS. ONE DIRECTION. ALWAYS SIMPLER.

I want to put this in its proper historical context, because I think the arc here is genuinely inspiring.

In the mid-1800s, physics had two apparently separate phenomena: electricity and magnetism. They seemed like completely different things. You had magnets — objects that attracted iron, aligned compasses, had north and south poles. And you had electricity — sparks, currents, lightning. Related? Maybe. The same thing? Surely not.

Then James Clerk Maxwell wrote down four equations. And those four equations showed that electricity and magnetism are not two things. They are two aspects of one thing: the electromagnetic field. Light — visible light, radio waves, X-rays — turned out to be this same field, oscillating. The unification revealed that reality was simpler and deeper than it had appeared.

Then, in the early twentieth century, Einstein noticed that space and time — again, apparently separate things — are also unified. Space is not a three-dimensional box in which time ticks away independently. They are four dimensions of a single entity: spacetime. The unification, again, revealed a deeper simplicity beneath apparent complexity.

"Maxwell unified electricity and magnetism. Einstein unified space and time. The substrate unifies spacetime and matter. Three steps. One direction. Toward a recognition that reality is simpler and more unified than it appears."

The pattern is consistent. Every time we have achieved a fundamental unification in physics, we have found that what appeared to be two different things was actually one thing, seen from two different angles or at two different scales. Unified Substrate Theory is proposing the next step in that sequence: spacetime and matter are not two different things. They are the same field in different states.

I have to be honest with you, as I always try to be. This is a proposal, not a finished theory. The full mathematical formalism — the equations that would let you make precise numerical predictions — is still being developed. The conceptual framework is clear. The physics intuition is strong. But the complete technical apparatus of a mature, published, peer-reviewed physical theory is not finished. That's real.

What IS finished are the predictions. The substrate model says: look for photon timing anomalies from gamma-ray bursts. Look for echoes in

gravitational waves. Test gravity's quantum nature in the lab. Look for evolution in dark energy. Those predictions are specific, falsifiable, and already being tested. That is what makes this science rather than speculation.

And I genuinely believe that within the next decade — maybe sooner — one or more of these experiments is going to return a result that changes everything. That has happened before. It will happen again. The question is only: what will the answer be?

BACK TO THE BEGINNING

I want to return to where we started. Those photons hitting your eyes. Those atoms in your body, vibrating. That chair, bending spacetime.

Here is what the substrate tells us about all of that.

Every atom in your body is a knot in the fabric of the universe. Not a metaphor. A real, topological structure in a real, physical field that fills all of existence. The light reaching your eyes right now is a ripple in that same fabric — not a particle flying through empty space, but the field itself, propagating a disturbance at the speed of light. The warmth you feel is the field transferring energy from one knot-cluster to another. The weight of you in that chair is the field curving, just slightly, around the mass you carry.

And every one of those phenomena — the knots, the ripples, the curves — is the substrate. One thing. Doing everything. All at once.

"You are not separate from the universe. You are a particularly organized region of it. You are the substrate, aware of itself."

I don't know if I've gotten every detail exactly right. I hold that with genuine humility. Science is a process of successive approximation — we get closer and closer to the truth, and anyone who tells you they've arrived at the final answer is someone to be cautious of. What I do believe, with everything I've thought and read and worked through, is that we are closer to reading the universe's native language than we have ever been before. The experiments are running. The data is coming in. And the language, underneath it all, might be simpler and more beautiful than any of us expected.

Thank you.

Dustin Lee is an independent science thinker and researcher based in Walstonburg, North Carolina. This essay is a public-facing explanation of Unified Substrate Theory (UST), written for a general audience. The scientific predictions referenced — including photon timing tests, gravitational wave echoes, quantum gravity table-top experiments, and dark energy evolution — are real ongoing areas of experimental investigation. The theoretical framework of UST is original work in active development. The author writes in the tradition of science popularizers who believe that the deepest ideas in physics belong to everyone, not just the specialist.